Internet of Things in Smart Agriculture

Research Report – Task 1: IoT in Real Life

1. Introduction

Agriculture must feed a growing population while coping with climate change, water scarcity and shrinking arable land. The Food and Agriculture Organization estimates that food production will need to rise by roughly 50% by 2050 compared with early-2010s levels [1]. Traditional farming, which relies on experience and uniform treatment of whole fields, struggles to meet this demand sustainably [2]. The Internet of Things (IoT) offers a practical response. By connecting sensors, controllers and software through the internet, IoT turns farms into data-driven systems, an approach commonly called smart agriculture or precision farming. This report explains how IoT works in farming, describes its main applications, and discusses its benefits, challenges and future direction.

2. How IoT Works on a Farm

A smart-farming system normally has four layers. The **sensing layer** consists of devices that measure soil moisture, pH, temperature, humidity, light, leaf wetness and water level, as well as GPS tags on animals and machinery [2][3]. The **network layer** carries this data to a gateway using technologies such as Wi-Fi, Bluetooth, ZigBee, LoRaWAN, NB-IoT or cellular networks. LoRaWAN is popular because it covers long distances at very low power, which suits large fields without mains electricity [4]. The **processing layer** stores and analyses the data on cloud or edge platforms, increasingly with machine learning. Finally, the **application layer** presents dashboards or mobile alerts to the farmer and sends commands to **actuators** such as valves, pumps and motors, closing the loop between measurement and action. Microcontrollers such as Arduino and ESP32 are widely used for low-cost prototypes of this architecture.

3. Key Applications

Precision irrigation. Irrigation is the clearest success story. Soil-moisture probes report water content at different depths, and the system opens a valve or starts a pump only when moisture falls below a set threshold. Combined with weather forecasts, this avoids both over-watering and crop stress, saving water and energy. One review reports that smart irrigation can cut agricultural water use by around 30–50% compared with conventional practice [2].

A simple prototype. Smart-farming ideas can be tested cheaply. A typical student prototype connects a capacitive soil-moisture sensor to an Arduino or ESP32 board, which compares the reading with a threshold and switches a small pump or motor through a transistor or relay. Indicator LEDs show whether the soil is wet or dry, and a potentiometer can set the threshold. Such circuits mirror, on a small scale, the sensing, decision and actuation loop used in commercial systems [2].

Soil and crop monitoring. Sensors for nutrients, pH and salinity guide variable-rate fertilisation, so each zone of a field receives only what it needs. Drones and satellites with multispectral cameras add a wider view, detecting plant stress, weeds or disease before they are visible to the eye [3][5].

Smart greenhouses. In greenhouses, IoT networks monitor temperature, humidity, CO_2 and light, and automatically control fans, heaters, shading and fogging to hold ideal growing conditions all year [3].

Livestock management. Wearable collars and ear tags track location, body temperature, activity and rumination. Changes in these signals can reveal illness or heat in an animal early, reducing losses and veterinary costs [4].

Pest and disease management. Connected traps and weather stations estimate pest pressure and disease risk, allowing targeted spraying instead of routine blanket application [2].

Supply chain and storage. Temperature and humidity loggers in warehouses and refrigerated trucks protect produce quality, while tagging and tracking supports traceability from farm to consumer [1][4].

4. Benefits

The main benefit is more efficient use of inputs. Studies report that sensor-guided irrigation, fertilisation and pest control optimise resource use, raise productivity and support sustainability [2][5]. Farmers gain real-time visibility of conditions, can act remotely and spend less labour on routine inspection. Environmental benefits follow: less fertiliser runoff, lower pesticide use and reduced water consumption. Over time, the accumulated data also allows better planning, yield prediction and informed decisions about crops and markets.

5. Challenges

Despite its promise, adoption faces real obstacles. Reviews consistently identify high initial investment, complicated data management, the need for technical skills, data security and privacy concerns, and poor connectivity in remote areas [2]. Small farmers in developing regions are especially affected by cost and limited digital literacy. Other issues include sensor calibration and drift in harsh outdoor conditions, battery life, and a lack of interoperability between devices from different vendors [4][5]. Questions about who owns farm data, and how it may be used by technology providers, also remain only partly resolved [1].

6. Future Trends

Several developments should lower these barriers. Low-power wide-area networks and cheaper sensors are reducing hardware cost, while edge computing lets devices make decisions locally when connectivity is weak. Integration of IoT with artificial intelligence is improving yield prediction, disease detection and automated decision-making, and robotics and autonomous tractors are beginning to act on sensor data directly [5]. Wider 5G coverage, shared cooperative platforms and open standards may also make smart farming affordable for smallholders.

7. Conclusion

IoT is changing agriculture from a practice based on estimates to one based on measurement. Through precision irrigation, crop and soil monitoring, smart greenhouses and livestock tracking, it helps farmers produce more with fewer resources. Cost, connectivity, security and skills gaps still limit its reach, but falling technology prices, supportive policy and training can address them. Smart agriculture is therefore likely to become a central tool for achieving food security and sustainable farming.

References

[1] Food and Agriculture Organization of the United Nations (2017). *The Future of Food and Agriculture: Trends and Challenges*. Rome: FAO.

[2] Mansoor, S., Iqbal, S., Popescu, S. M., Kim, S. L., Chung, Y. S. and Baek, J.-H. (2025). Integration of smart sensors and IoT in precision agriculture: trends, challenges and future prospectives. *Frontiers in Plant Science*, 16, 1587869. DOI: 10.3389/fpls.2025.1587869.

[3] Soussi, A., Zero, E., Sacile, R., Trinchero, D. and Fossa, M. (2024). Smart sensors and smart data for precision agriculture: a review. *Sensors*, 24(8), 2647. DOI: 10.3390/s24082647.

[4] Ayaz, M., Ammad-Uddin, M., Sharif, Z., Mansour, A. and Aggoune, E.-H. M. (2019). Internet-of-Things (IoT)-based smart agriculture: toward making the fields talk. *IEEE Access*, 7, 129551–129583.

[5] Manono, B. O. and Mwami, B. M. (2026). Precision farming with smart sensors: current state, challenges and future outlook. *Sensors*, 26(3).